Financial Management

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2.2: Capital Budgeting

Agenda

- Capital budgeting concept
- Capital budgeting techniques
- Investment project ranking
- Capital rationing
- Risk analysis in capital budgeting

Capital Budgeting

- " Capital Budgeting is the firm's decision to invest its current funds most efficiently in the long-term assets in anticipation of an expected flow of benefits over a series of years."
- It is the process of identifying, analyzing and selecting investment projects whose returns are expected to extend beyond one year.
- It influence the firm's growth in the long run
- It affects the risk of the firm
- It involves commitment of large amount of funds
- it is irreversible or reversible at substantial loss
- It is the most difficult decision to make

Capital Budgeting

- Capital Budgeting involves buying expensive assets for long term use. The purpose may be
  - Expansion of existing business
  - Expansion of new business
  - Replacement and modernization
- Investment decision can be classified as
  - Mutually exclusive investments
  - Independent investments
  - Mandatory investments
  - Special projects

Capital Budgeting Decisions in AI Driven Firms

In AI-driven firms, capital budgeting decisions often involve:

- **AI infrastructure (cloud, GPUs, data centers).**
  –Eg. Investing in NVIDIA GPUs for GenAI model training
- **AI product development (ML models, platforms).**
  –Eg. Developing a proprietary recommendation engine
  –Implementing AI-powered fraud detection in fintech
- **Automation & intelligent systems**
  –Eg: Automating customer support using chatbots

- These decisions affect scalability, data advantage, and competitive edge.

Capital Budgeting Evaluation Criteria

Three steps are involved in the evaluation of an investment:

1. Estimation of cash flows
2. Estimation of the required rate of return
3. Application of a decision rule for making the choice

Estimation of Cash Flow

Cash Flow = Profit + Depreciation – Capital Expenditure

OR

FCF = operating profit + depreciation – tax – additional investment

OR

FCF = Net operating profit after tax – Net investment

Cash Inflow = Profit + Depreciation

Cash Outflow = Capital Expenditure

Estimation of Cash Flow: Illustration- Manufacturing

A company is considering expanding its production. It can go in either for an automatic machine costing Rs.224000 with an estimated life of 5 years or an ordinary maching costing Rs.60000 having an estimated life of 8 years. Tax rate is 50%.The annual sales and costs are estimated as follows.

	Automatic Machine (Rs.)	**Ordinary Machine (Rs.)**
Sales	150000	150000
Raw material cost	50000	50000
Cost of labour	12000	60000
Variable overheads	24000	20000

Calculate cash inflows for both the machines.

Estimation of Cash Flow Illustration: AI Chatbot

AI Chatbot Implementation

Initial investment: Rs. 50 lakhs

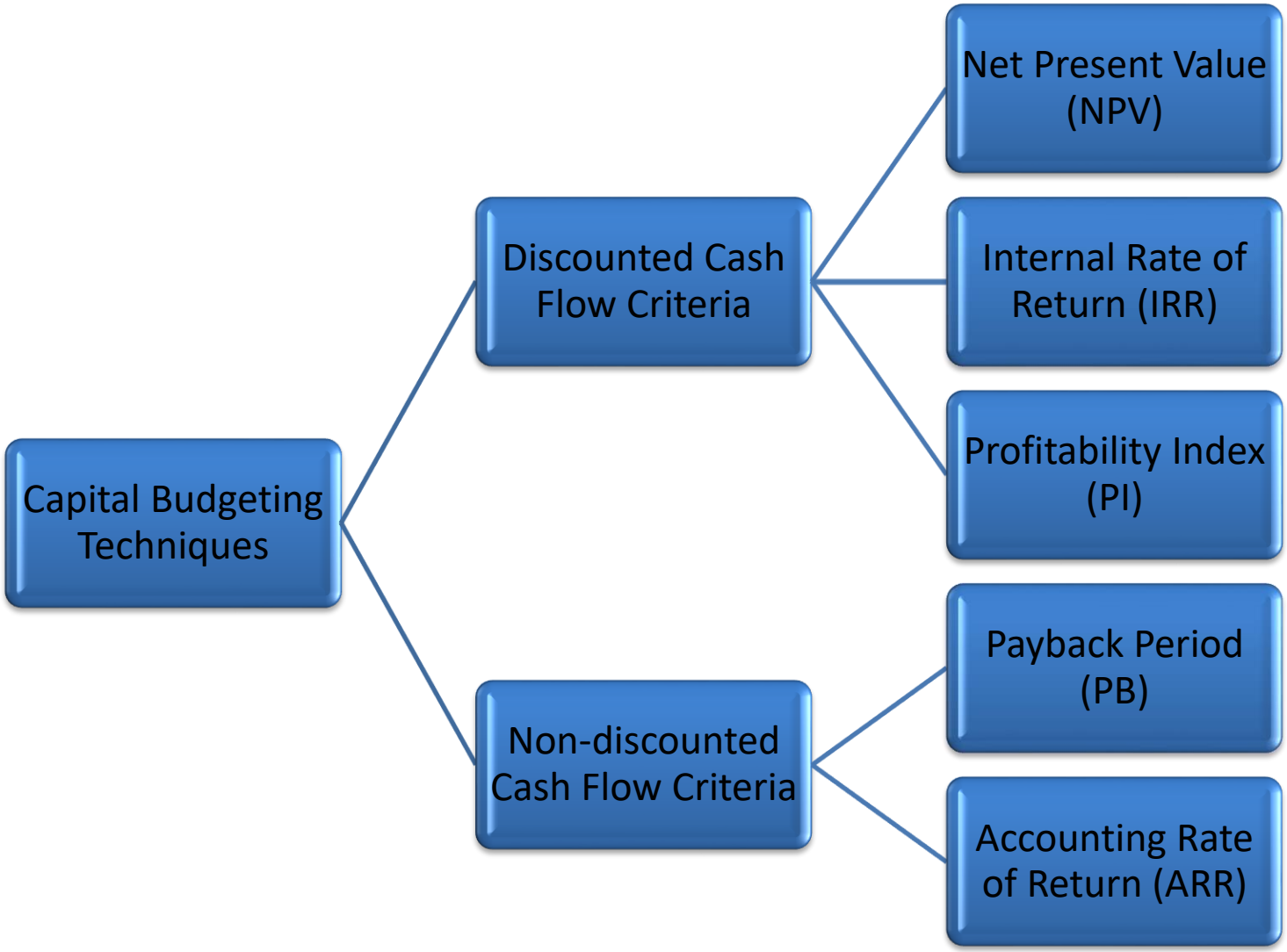
Annual savings from reduced manpower: Rs.18 lakhs

Additional revenue: Rs. 7 lakhs

Useful life: 5 years

Annual Cash Inflow = Savings + Incremental Revenue

Capital Budgeting Techniques



Net Present Value (NPV)

- NPV is based on the economic principle, “Time Value of Money”: Cash flows of the project received earlier has more value than the cash flows received later.
- It postulates that cash flows arising at different time periods differ in value and are comparable only when their equivalents- present values – are found out.

$$\textbf{NPV} = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} - C_o$$

Where,

C_t = net cash inflows in year 1,2,3,….n

k = opportunity cost of capital (assumed to be known and constant)

C_o = initial cost of the investment

n = expected life of the investment

Net Present Value (NPV)

Acceptance Rule

- Accept if NPV > 0
- Reject if NPV < 0
- May be accepted if NPV = 0

Net Present Value (NPV): Illustration 1

Assume that project X costs Rs.25000 now and is expected to generate year-end cash inflows of Rs.9000, Rs.8000, Rs.7000, Rs.6000 and Rs.5000 in year 1 through 5. The opportunity cost of the capital may be assumed to be 10%. Decides on the projects based on the NPV method.

$$\textbf{NPV} = \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} - C_o$$

$$\text{NPV} = \left[\frac{Rs.9000}{(1+0.10)^1} + \frac{Rs.8000}{(1+0.10)^2} + \frac{Rs.7000}{(1+0.10)^3} + \frac{Rs.6000}{(1+0.10)^4} + \frac{Rs.5000}{(1+0.10)^5}\right] - \text{Rs. } 25000$$

NPV = Rs.27250 – Rs.25000 = Rs.2250

Accept as NPV > 0

NPV: Illustration 2

AI Startup plans to build a recommendation engine.

Initial Investment: Rs.1 crore

Expected Cash Inflows (Rs. lakhs):

Year 1–5: 20, 25, 30, 35, 40

Cost of Capital: 12%

NPV: Illustration 3

Assume that project X costs Rs.5000000 now and is expected to generate year-end cash inflows of Rs.100000, Rs.1500000, Rs.1750000, Rs.2000000 and Rs.2500000 in year 1 through 5. The opportunity cost of the capital may be assumed to be 8%. Decides on the projects based on the NPV method.

Internal Rate of Return (IRR)

- IRR is based on the economic principle, “Time Value of Money”: Cash flows of the project received earlier has more value than the cash flows received later.
- IRR is the rate at which project will have a zero NPV

$$\textbf{IRR} = \sum_{t=1}^{n} \frac{C_t}{(1+r)^t} - C_o = 0$$

Where,

C_t = net cash inflows in year 1,2,3,….n

r = rate of return (to be determined using trial and error method)

C_o = initial cash outflow

n = expected life of the investment

Internal Rate of Return (IRR)

Using Trail and Error Method, following formula can be used to find out a close approximation of the rate of return

$$\textbf{IRR} = \textbf{LDF \%} + \left(\Delta\mathbf{DF}\ \frac{\mathbf{PVLDF - COF}}{\mathbf{PVLDF - PVHDF}}\right)$$

Where,

LDF = Lower discount factor

ΔDF = Difference between high discounting factor and low discounting factor

PVLDF = PV of cash inflows at low discounting factor

PVHDF = PV of cash inflows at high discounting factor

COF = Initial cash outflow

Internal Rate of Return (IRR)

Acceptance Rule

- Accept if $r > k$
- Reject if $r < k$
- May be accepted if $r = k$

Internal Rate of Return (IRR): Illustration 1

A project costs Rs.16000 and is expected to generate cash inflows of Rs.8000, Rs.7000 and Rs.6000 at the end of each year for next 3 years. Decides on the project using IRR method.

IRR = $\sum_{t=1}^{n} \frac{C_t}{(1+r)^t} - C_o = 0$

Use Trial and Error Method.

Calculate NPV at different rate to reach to zero (NPV =0).

Start with k = 20%

And then put a formula to find out IRR

IRR = LDF % + $\left(\Delta \mathrm{DF} \frac{\mathrm{PVLDF} - \mathrm{COF}}{\mathrm{PVLDF} - \mathrm{PVHDF}}\right)$

IRR: Illustration 2

Project: AI-based warehouse automation

Initial Outlay: Rs.80 lakhs

Cash Inflows (Rs. lakhs): 20, 25, 30, 30

Cost of capital : 11%

Profitability Index (PI)

The method is also called benefit cost ratio.

$$\text{Profitability Index} = \frac{\sum_{t=1}^{n} \frac{C_t}{(1+r)^t}}{C_0}$$

Acceptance Rule

- Accept if PI> 1
- Reject if PI < 1
- May be accepted if PI = 1

PI: Illustration

The initial cash outlay of a project is Rs. 100000 and it can generate cash inflow of Rs. 40000, Rs.30000, Rs.50000 and Rs.20000 in year 1 through 4.. Assume a 10 per cent rate of discount and evaluate the project using profitability index method.

PV = Rs. 40000($PVIF_{1,0.10}$) + Rs. 30000($PVIF_{2,0.10}$) + Rs. 50000($PVIF_{3,0.10}$) + Rs. 00000($PVIF_{4,0.10}$)

PV = Rs 112350

Profitability Index = $\frac{\sum_{t=1}^{n}\frac{C_t}{(1+r)^t}}{C_0}$ = $\frac{Rs.112350}{Rs.100000}$ = 1.235 > 1

Accept the project

PI: Illustration 2

Company migrates to AI cloud infrastructure

PV of future cash inflows: Rs.1.5 crore

Initial investment: Rs.1.2 crore

Payback Period

- PB is a non-discounted method of capital budgeting
- It is defined as the number of years required o recover the original cash outlay invested in a project.
- If the project generates constant annual cash inflows, he payback period can be computed as

$$\textbf{Payback} = \frac{\textbf{Initail Investment}}{\textbf{Annual Cash Inflow}} = \frac{C_o}{C}$$

- In case of unequal cash inflow, the payback period can be found out by adding up the cash inflows until the total initial cash outlay.

Payback Period

Acceptance Rule

- Accept if PB < Standard PB
- Reject if PB > Standard PB

Limitations:

- A major shortcoming of the conventional payback period is that it does not taken into account the time value of money.

Payback Period: Illustration 1

Assume that a project requires an outlay of Rs. 50000 and yields an annual cash inflow of Rs.12500 for 7 years. Calculate payback period to decide on the project.

$$\textbf{Payback} = \frac{\textbf{Initail Investment}}{\textbf{Annual Cash Inflow}} = \frac{C_o}{C}$$

$$\text{Payback} = \frac{Rs.50000}{Rs.12500} = 4 \text{ years}$$

Payback Period: Illustration 2

Suppose that a project requires a cash outlay of Rs.20000, and generates cash inflows of Rs.8000, Rs.7000, Rs.4000 and Rs.3000 during he next 4 years. Decides on the project using payback period method.

To find out payback, add up the cash inflows and find when it recovers the original outlay.

1 – 3 years cash inflows = Rs.8000+Rs.7000+Rs.4000

= Rs.19000

To recover original outlay i.e. Rs.20000, Rs.1000 required in 4th year.

So the time required to recover Rs.1000 in the 4th year =

= (Rs.1000/Rs.3000) X 12 months = 4 months

Therefore, the payback period id 3 years and 4 months which is less than the standard 4 years. So project can be accepted.

Discounted Payback Period

- Cash flows are first converted into their present values and then added to ascertain the period of time required to recover the initial outlay on the project.

Illustration: Discounting factor is 10%

Year	Cashflow of A (Rs.)	Cash Flow of B (Rs.)
0	(100000)	(100000)
1	50000	20000
2	30000	20000
3	20000	20000
4	10000	40000
5	10000	50000
6	-	60000

Discounted Payback: Illustration 2

AI-based demand forecasting system

Initial Cost: Rs.40 lakhs

Annual Cash Inflows (Rs. lakhs): 10, 12, 15

Cost of capital: 10.5%

MINI CASE

Aman Ltd. is a leading manufacturer of automotive components. It supplies to the original equipments manufacturers as well as the replacement market. Its projects typically have a short life as a introduces new models periodically.

You have recently joined Aman Ltd. As a financial analyst to Ravi Sharma, the CFO of the company. He has provided you the following information about three projects, A, B, and C that are being considered by the Executive Committee of Aman Ltd.

- Project A is an extension of an existing line. Its cash flow will decrease over time.
- Project B involves a new product. Building its market will take some time and hence its cash flow will increase over time.
- Project C is concerned with sponsoring a pavilion at a Trade Fair. It will entail a cost initially which will be followed by a huge benefit for one year. However, in the year following that some cost will be incurred to raze the pavilion.

MINI CASE

The expected net cash flows of the three projects are as follows:

Year	Project A (Rs.)	Project B (Rs.)	Project C (Rs.)
0	(15000)	(15000)	(15000)
1	11000	3500	42000
2	7000	8000	(4000)
3	4800	13000	

Ravi Sharma believes that all the three projects have risk characteristics similar to the average risk of the firm and hence the firm's cost of capital, viz. 12 percent, will apply to them.

MINI CASE

You have been asked to prepare a report for the executive committee, covering the following:

a. Find the payback period and the discounted payback period of Projects A and B.
b. Calculate NPV of all the projects.
c. Calculate IRR for projects A, B, and C.

Practice Problem: NPV

The firm has two investment opportunities, each costing Rs.100000 and each having an expected cash inflow as shown below:

Year	1	2	3	4
Project A (Rs.)	50000	40000	30000	10000
Project B (Rs.)	20000	40000	50000	60000

After giving the consideration to the risk criteria in each project the management has decided that project A should be evaluated at 10% cost of capital and project B, a risk project with a 15% cost of capital.

Calculate the NPV and suggest the course of action for the management if

(a) Both the project are independent

(b) Both are mutually exclusive

Practice Problem : IRR

Flipkart is considering the following two proposals for its facilities centers.

Year	0	1	2	3
Project P (Rs.)	2500000	500000	500000	2564000
Project Q (Rs.)	2800000	1267200	1267200	1260200

Evaluate and suggest the company which project should be selected using IRR method. The opportunity cost of capital is 12%.

Practice Problem : PI

The cash flows associated with an investment are given below. Calculate the benefit cost ratio of this investment. Discounting rate is 10%

Year	Cash Flow (in Rs.)
0	(100000)
1	25000
2	40000
3	50000
4	40000
5	30000

Practice Problem : PB

A project expected cash flows are as follows:

Year	0	1	2	3	4	5
Cash Inflows (Rs.)	(50000)	10000	15000	20000	25000	20000

Calculate Payback period

THANK YOU!!!